

Git for Science

A substrate for reproducible, compounding scientific knowledge

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Git did not just store files. It stored the *repository*: a structured, content-addressed object with identity and lineage, of which the working files are one view. That object is why merging, provenance, and collaboration became mechanical, and why a planet's worth of code could accumulate in one shared form. Science is still pre-git. We ship the paper, the printout of a result, and throw the object away. This is an attempt to store the object: what a result *is* (a typed physical quantity, its formula, its lineage, and the code that produced it as one representation of it), so that results become reproducible by construction, knowledge compounds instead of evaporating, and an AI agent has a semantic place to act.

The creed

1. **The unit of scientific knowledge is meaning, not text.**
2. **Physics is the invariant. A code is one representation of it.**
3. **A scientific result is a structured object with identity and lineage, not a number in a table.**
4. **A result you cannot re-run is a claim, not yet knowledge.**
5. **Units, gauge, and lineage are part of a result, not footnotes to it.**
6. **Knowledge should compound, not evaporate into scripts.**
7. **Correctness should be structural, checked by the substrate, not re-derived by trust.**
8. **Verification must scale with compute, not with reviewers.**
9. **The native substrate for scientific AI is semantic, not token text.**